

BOSTON TEA PARTY 2.0

Project Plan · February 27 – April 10, 2026

Ockert · Di Mise · Ma · Khandelwal

§ I — Team Roles & Responsibilities

Jonathan Ockert, CPA — Constraint Engine & Domain Lead Owns the tax logic: defines constraint rules from the IRC, curates benchmark test scenarios, and validates all system outputs for domain correctness. His CPA background makes him the natural domain authority for the project.

Francesco Di Mise — Verification Layer & Evaluation Lead Builds the faithfulness checking system, designs and runs the evaluation benchmarks (constraint accuracy, RAG faithfulness, hallucination detection), manages the baseline LLM comparison, and leads the user comprehension study.

Anthony Ma — Planning Agent & Integration Lead Develops the agentic orchestration layer: follow-up question logic, replanning when a user's fact pattern changes, and wiring all components together. Also responsible for the demo UI. Acts as the connective tissue between all system components.

Ayushman Khandelwal — RAG Pipeline & Data Engineer Builds the retrieval system end-to-end using a hybrid approach: PageIndex-style tree indexing for hierarchical sources (IRC, IRS Publications) and lightweight retrieval for narrative sources (Tax Court opinions). Responsible for document parsing, tree construction, and retrieval tuning.

§ II — Retrieval Strategy: Vectorless + Hybrid RAG

A key architectural decision for this project is to move away from a traditional vector-based RAG pipeline in favor of a hybrid retrieval approach grounded in structural reasoning.

Why vectorless for tax documents? The IRC, IRS Publications, and Tax Court opinions are not unstructured blobs of text. The IRC is organized by title, chapter, subchapter, and section. IRS Publications are structured with numbered headings, worked examples, and tax-year editions. Traditional vector RAG destroys this hierarchy by chunking documents into flat fragments and relying on semantic similarity, which often retrieves text that sounds related but isn't actually relevant to the legal question being asked.

PageIndex and tree-based retrieval. PageIndex is an open-source framework (VectifyAI) that builds a hierarchical tree index from documents and uses LLM reasoning to navigate that tree — mimicking how a human expert scans a table of contents, drills into the right chapter, and locates the relevant subsection. On the FinanceBench benchmark (structurally similar to tax filings), this approach achieved 98.7% accuracy compared to roughly 50% for traditional vector RAG.

Our hybrid approach. We will use tree-based reasoning retrieval for the IRC and IRS Publications, where the hierarchical structure is deep and well-defined. For Tax Court opinions, which are more narrative and prose-based, we will use a simpler keyword or BM25-based retrieval approach. This hybrid design gives us both accuracy on structured sources and flexibility on narrative ones, and also provides a natural comparison point for our evaluation section.

Tradeoffs. Tree-based retrieval adds LLM reasoning calls at query time, increasing latency and cost per query compared to vector lookup. For a demo-scale system this is acceptable, but worth acknowledging. The hybrid approach mitigates this by reserving the heavier reasoning retrieval for the sources that benefit most from it.

§ III — Timeline Overview

The project spans six weeks from February 27 to April 10. The critical integration checkpoint falls at the end of Week 3 (March 19), when all components must work together end-to-end for the first time. If the team is behind at that point, Week 4 should shift to triage rather than polish.

Week	Dates	Phase
1	Feb 27 – Mar 5	Foundation & Data Collection
2	Mar 6 – Mar 12	Core Component Prototypes
3	Mar 13 – Mar 19	Integration & Verification
4	Mar 20 – Mar 26	Hardening & Edge Cases
5	Mar 27 – Apr 2	Evaluation & User Study
6	Apr 3 – Apr 10	Demo Prep & Final Polish

§ IV — Week-by-Week Plan

Week 1: Foundation & Data Collection (Feb 27 – Mar 5)

Milestone: Raw data collected, constraint schema drafted, dev environment set up, document structure analysis complete.

All: Set up shared repository, development environment, and agree on the tech stack. Key decisions: LLM provider, PageIndex or custom tree indexer, BM25 library (e.g., rank-bm25 or Elasticsearch), and orchestration framework (e.g., LangChain or LlamaIndex).

Jonathan: Draft the constraint rule schema by mapping IRC eligibility rules to structured logic — filing status to valid deductions, income thresholds, and documentation requirements. Identify 10–15 core tax scenarios for benchmarking, covering varied filing statuses, multi-state work, and itemized vs. standard deductions.

Ayushman: Source and download IRC sections, IRS Publications (Pub. 17, 501, 526, etc.), and sample U.S. Tax Court opinions. Analyze the structural hierarchy of each source type — map heading levels, section numbering, and nesting patterns in the IRC and IRS Pubs. Evaluate PageIndex for tree construction or determine if custom parsing is needed for tax-specific document structures.

Anthony: Design the intake schema defining what structured fields the system collects from users (filing status, income sources, states of employment, deduction preferences). Sketch the agent orchestration flow as a state machine or directed acyclic graph for conversation planning.

Francesco: Research faithfulness evaluation methods such as RAGAS and citation F1 scoring, along with hallucination detection approaches. Draft the evaluation plan document covering metrics, benchmark format, baseline LLM comparison design, and a plan to compare tree-based vs. keyword retrieval across document types.

Week 2: Core Component Prototypes (Mar 6 – Mar 12)

Milestone: Each component works independently — constraint engine returns valid/invalid paths, tree-based retrieval navigates IRC/IRS Pubs, BM25 retrieves from Tax Court opinions, planning agent asks follow-ups.

Jonathan: Implement constraint engine v1 with deterministic rules for filing status, standard vs. itemized deductions, and charitable deduction eligibility. Write unit tests for the constraint logic using the curated scenarios.

Ayushman: Build the tree index for IRC sections and IRS Publications — parse document hierarchy, construct the tree structure, and implement LLM-based tree navigation for retrieval. Set up BM25 or keyword-based retrieval for Tax Court opinions as the secondary retrieval path. Ensure both retrieval paths return source metadata (section numbers, publication year, case name).

Anthony: Implement planning agent v1 that takes a user's fact pattern, identifies unresolved constraints, and generates follow-up questions. Define the interface contracts between components (constraint engine ↔ agent ↔ retrieval layer), including a unified retrieval interface that abstracts over the two retrieval strategies.

Francesco: Build the baseline by running a vanilla LLM (no retrieval, no constraints) against the same benchmark scenarios. Set up the evaluation harness with automated scoring for constraint accuracy and citation grounding rate.

Week 3: Integration & Verification (Mar 13 – Mar 19)

Milestone: End-to-end pipeline works — intake → constraints → agent → retrieval (tree + BM25) → verified output.

▮ **CRITICAL CHECKPOINT — This is the make-or-break week.** All four components must talk to each other for the first time. If the pipeline doesn't work end-to-end by March 19, the team should meet to triage and reprioritize Week 4.

Anthony: Wire all components into the end-to-end pipeline so that a user fact pattern flows through the full system. The retrieval layer should route queries to tree-based search for IRC/IRS Pubs and BM25 for Tax Court opinions based on query type or source relevance. Implement replanning logic so that when a user changes a fact, the system re-evaluates all constraints and updates the session.

Francesco: Build verification layer v1 to check that generated explanations actually cite retrieved sources and flag unsupported claims. Implement tax-year accuracy checking to ensure correct publication editions are being retrieved.

Jonathan: Expand constraint rules to cover multi-state scenarios and edge cases. Review integrated system outputs against expert-labeled ground truth and identify failure modes.

Ayushman: Tune tree-based retrieval — test different tree depths, adjust the LLM reasoning prompts for node selection, and evaluate whether the tree structure captures the right granularity for tax questions. Tune BM25 parameters for Tax Court opinion retrieval. Compare retrieval quality across the two strategies.

Week 4: Hardening & Edge Cases (Mar 20 – Mar 26)

Milestone: System handles all benchmark scenarios correctly; verification layer catches hallucinations; both retrieval paths perform well.

Jonathan: Finalize constraint coverage to ensure all 10–15 benchmark scenarios produce correct eligibility decisions. Document all constraint rules and their corresponding IRC source references.

Ayushman: Optimize retrieval performance — improve tree navigation prompts, reduce unnecessary LLM calls in the tree search, and handle edge cases where documents don't parse cleanly. Ensure all three data sources (IRC, IRS Pubs, Tax Court) are retrievable and properly attributed.

Anthony: Harden the agent conversation flow to handle ambiguous inputs, conflicting facts, and missing information gracefully. Build a simple demo UI (Streamlit or Gradio) for interactive walkthroughs.

Francesco: Run the full evaluation suite covering constraint accuracy, retrieval faithfulness, and hallucination rate vs. the baseline. Include a comparison of tree-based retrieval vs. BM25 across document types to quantify the benefit of structural reasoning. Collect initial results and identify the biggest gaps to fix in Week 5.

Week 5: Evaluation & User Study (Mar 27 – Apr 2)

Milestone: Quantitative results finalized; user comprehension study complete; retrieval strategy comparison documented.

Francesco: Run the user comprehension study with 5–8 participants, using pre/post comprehension scoring on realistic tax scenarios. Compile all quantitative results: constraint accuracy, citation grounding rate, hallucination rate, baseline comparison, and tree-based vs. BM25 retrieval comparison.

Jonathan: Perform a final domain-accuracy QA pass on all system outputs. Write up the constraint engine methodology section for the final presentation.

Ayushman: Fix any remaining retrieval issues surfaced during evaluation. Write up the data pipeline and retrieval strategy section, covering the rationale for the hybrid approach and the results of the retrieval comparison.

Anthony: Polish the demo UI to ensure it clearly showcases the full pipeline for the presentation. Write up the architecture and agent design section.

Week 6: Demo Prep & Final Polish (Apr 3 – Apr 10)

Milestone: 🎯 April 10 — Final presentation and live demo.

All: Build the final presentation by updating the existing slide deck with results, architecture diagrams, and demo screenshots. Rehearse the demo end-to-end and assign who presents each section. Prepare 2–3 compelling demo scenarios that showcase constraints, tree-based retrieval with citations, replanning, and verification in action.

Anthony: Ensure the demo environment is stable and prepare a backup recorded walkthrough in case of live issues.

Francesco: Prepare results slides with clear visuals — charts comparing the system vs. the vanilla LLM baseline, and tree-based vs. BM25 retrieval across document types.

Jonathan: Prepare to field domain-specific Q&A during the presentation.

Ayushman: Prepare technical deep-dive backup slides on the hybrid retrieval strategy (PageIndex tree indexing vs. BM25) in case of audience questions.

§ V — Key Risks & Mitigations

Integration delays (Week 3): The biggest risk is that independently-built components don't interface cleanly. Mitigation: define interface contracts in Week 2 before integration begins, and treat March 19 as a hard internal deadline.

Tree index quality: If the IRC or IRS Pub documents don't parse into clean hierarchical trees, the reasoning-based retrieval will underperform. Mitigation: Ayushman should analyze document structure in Week 1 and flag parsing challenges early. Have a fallback plan to use BM25 across all sources if tree indexing proves too unreliable for the timeline.

LLM cost and latency: Tree-based retrieval requires LLM calls at query time, which adds cost and latency. Mitigation: scope tree retrieval to the sources that benefit most (IRC, IRS Pubs), use BM25 for Tax Court, and cache tree navigation results for repeated query patterns.

Constraint coverage gaps: Tax law has countless edge cases. Mitigation: scope the constraint engine to the 10–15 curated scenarios rather than trying to cover the full IRC. Be explicit about scope in the demo.

User study recruitment: Finding 5–8 participants in Week 5 requires advance planning. Mitigation: begin recruiting in Week 3 so participants are confirmed before the study begins.

§ VI — Definition of Done (April 10)

The project is complete when the team can demonstrate: hard eligibility rules enforced correctly by the constraint engine, every generated claim backed by a cited source retrieved through tree-based or keyword search, coherent replanning when a user's fact pattern changes mid-session, and quantitative evaluation results showing concrete improvement over a vanilla LLM baseline. The evaluation should also show the benefit of structural tree-based retrieval over flat retrieval for hierarchical tax documents. Users who interact with the system should understand more about their tax situation than when they started.
